Ionatan Kuperwajs

Curriculum Vitae

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EMPLOYMENT

Princeton University, Princeton, NJ — 2024-Present
Postdoctoral Research Associate
Department of Computer Science
Advisor: Thomas L. Griffiths

EDUCATION

New York University, New York, NY — 2018-2024
Ph.D. in Neural Science
Thesis: Cognitive mechanisms of complex planning
Advisor: Wei Ji Ma

Macalester College, St. Paul, MN — 2014-2018
B.A. in Neuroscience, Computer Science, and Mathematics
Honors in Mathematics, Magna Cum Laude
Advisor: Andrew Beveridge

AWARDS AND FELLOWSHIPS

New York University Graduate School of Arts and Sciences Dean's Outstanding Dissertation Award	2024
National Science Foundation Graduate Research Fellowship	2020-2023
Henry Mitchell McCracken Fellowship	2018
Phi Beta Kappa National Honor Society Member	2018
Macalester College Neuroscience Department Outstanding Graduate Award	2018
Macalester College Dean's List	2014-2018
Macalester College DeWitt Wallace Distinguished Scholar	2014

PUBLICATIONS

Preprints

I Kuperwajs, EM Russek, L Schut, Y Sagiv, WJ Ma, and TL Griffiths (2026). Time pressure modulates resource-rational planning in online chess. *PsyArXiv.*

I Kuperwajs and WJ Ma (2026). Learning progress shapes people's voluntary engagement. *PsyArXiv.*

I Ben-Artzi, EM Russek, **I Kuperwajs**, N Shahar, M Mattar, WJ Ma, and TL Griffiths (2026). People make history-dependent decisions in Markovian environments. *PsyArXiv.*

I Kuperwajs, B van Opheusden, EM Russek, and TL Griffiths (2024). Learning from rewards and social information in naturalistic strategic behavior. *PsyArXiv.*

I Kuperwajs, MK Ho, and WJ Ma (2024). Heuristics for meta-planning from a normative model of information search. *PsyArXiv.*

Journal articles

I Kuperwajs, EM Russek, MG Mattar, WJ Ma, and TL Griffiths (2026). Looking deeper into the algorithms underlying human planning. *Trends in Cognitive Sciences.*

I Kuperwajs, HH Schütt, and WJ Ma (2023). Using deep neural networks as a guide for modeling human planning. *Scientific Reports.*

B van Opheusden, **I Kuperwajs**, G Galbiati, Z Bnaya, Y Li, and WJ Ma (2023). Expertise increases planning depth in human gameplay. *Nature.*

Conference proceedings

I Lee, EM Russek, **I Kuperwajs**, and TL Griffiths (2026). Identifying concepts used by human-like neural network chess engines. *Proceedings of the 48th Annual Meeting of the Cognitive Science Society.*

KM Collins, CE Zhang, G Todd, L Ying, MB da Costa, R Liu, P Sharma, A Weller, **I Kuperwajs**, L Wong, JB Tenenbaum, and TL Griffiths (2026). Evaluating language models' evaluations of games. *Proceedings of the 14th International Conference on Learning Representations.*

I Kuperwajs, EM Russek, L Schut, Y Sagiv, MG Mattar, WJ Ma, and TL Griffiths (2025). Exploring resource-rational planning under time pressure in online chess. *Proceedings of the 47th Annual Meeting of the Cognitive Science Society.*

L Wong, T Mills, **I Kuperwajs**, KM Collins, and TL Griffiths (2025). Meta-reasoning: deciding which game to play, which problem to solve, and when to quit. *Proceedings of the 47th Annual Meeting of the Cognitive Science Society.*

I Kuperwajs, HH Schütt, and WJ Ma (2022). Improving a model of human planning via large-scale data and deep neural networks. *Proceedings of the 44th Annual Meeting of the Cognitive Science Society.*

I Kuperwajs and WJ Ma (2022). A joint analysis of dropout and learning functions in human decision-making with massive online data. *Proceedings of the 44th Annual Meeting of the Cognitive Science Society.*

I Kuperwajs and WJ Ma (2021). Planning to plan: a Bayesian model for optimizing the depth of decision tree search. *Proceedings of the 43rd Annual Meeting of the Cognitive Science Society.*

I Kuperwajs, B van Opheusden, and WJ Ma (2019). Prospective planning and retrospective learning in a large-scale combinatorial game. *Cognitive Computational Neuroscience.*

SUMMER SCHOOLS AND INTERNSHIPS

Simons Computational Neuroscience Imbizo, Cape Town, South Africa	2018
Howard Hughes Medical Institute Janelia Undergraduate Scholars Program, Ashburn, VA	2017
New York University Center for Neural Science National Science Foundation REU, New York, NY	2016

INVITED TALKS

Department of Psychology, Tufts University	2025
Reinforcement Learning and Decision Making (Workshop), Trinity College Dublin	2025
Parallel Distributed Processing Seminar, Princeton University	2024
Computational Cognitive Neuroscience Lab, University of California, Berkeley	2023
Computational Cognitive Science Lab, Princeton University	2023
Cognitive Science Society, Toronto, Canada (2 talks)	2022
Cognitive Science Society, University of Vienna	2021
Center for Neural Science Seminar, New York University	2020
Concepts and Categories Seminar, New York University	2019

Artificial and Biological Computation Lab, New York University 2019
Sensorimotor Learning Group, Columbia University 2019

CONFERENCE POSTERS

Cognitive Science Society, Rio de Janeiro, Brazil 2026
International Conference on Learning Representations, Rio de Janeiro, Brazil 2026
Neural Information Processing Systems (Workshop), San Diego, United States 2025
Cognitive Science Society, San Francisco, United States (2 posters) 2025
Association for the Advancement of Artificial Intelligence (Bridge), Vancouver, Canada 2024
Cognitive Computational Neuroscience, Oxford University 2023
Minds, Brains, and Machines, New York University 2023
Reinforcement Learning and Decision Making, Brown University (3 posters) 2022
Scaling Cognitive Science, Princeton University 2019
Cognitive Computational Neuroscience, Technical University of Berlin 2019

TEACHING

New York University, Teaching Assistant
NEURL-GA 2201: Mathematical Tools for Neural and Cognitive Science F 19

Macalester College, Teaching Assistant
COMP 221: Algorithm Design and Analysis F 17, S 18
PSYC 180: Brain, Mind, and Behavior F 16
COMP 123: Core Concepts in Computer Science S 16, F 16

University of California, San Diego, Guest Lecturer
COGS 11D: Computational Models of Cognition and Behavior F 26

MENTORSHIP

Graduate students

Ido Ben-Artzi, Tel Aviv University (Neuroscience) 2025-Present
Lisa Schut, University of Oxford (Computer Science) 2024-Present

Undergraduate students

Kathryn McGregor, Princeton University (Computer Science) 2026-Present
Natalia Dice, Princeton University (Computer Science) 2026-Present
Anany Kotawala, Princeton University (Operations Research and Financial Engineering) 2026-Present
Issac Li, Princeton University (Computer Science) 2025-Present
Kirill Sirik, Princeton University (Operations Research and Financial Engineering) 2025
Brandon Lee, Princeton University (Computer Science) 2025

SERVICE

Ad hoc reviewing

Cognitive Computational Neuroscience
Cognitive Science
Cognitive Science Society
Communications Psychology
Journal of Expertise
Nature Communications
Nature Human Behaviour

Philosophical Transactions of the Royal Society A
Science Advances
Scientific Reports

Workshop organization

Meta-reasoning: deciding which game to play, which problem to solve, and when to quit 2025
Co-organizer, Cognitive Science Society, San Francisco, United States

Chess as a bridge between human cognition and artificial intelligence 2025
Co-organizer, Reinforcement Learning and Decision Making, Trinity College Dublin

ADVOCACY AND OUTREACH

Member, Princeton University Psychology Department DEI Committee 2025-Present
President, Scientist Action and Advocacy Network (ScAAN) 2018-2023
Workshop on the Climate Crisis, Cognitive Computational Neuroscience, Oxford University 2023
Workshop on Environmental Justice, Ocean Sciences Meeting 2022
Workshop on Evidence-Based Advocacy, American Geophysical Union, New Orleans, LA 2021
Workshop on Science Activism, Rockefeller University 2021
Panel on Science Activism, Growing Up in Science 2020

REFERENCES

Thomas L. Griffiths
Professor, Princeton University
Departments of Psychology and Computer Science
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Wei Ji Ma
Professor, New York University
Center for Neural Science and Department of Psychology
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Heiko H. Schütt
Associate Professor, University of Luxembourg
Department of Behavioural and Cognitive Sciences
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Mark K. Ho
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Department of Psychology
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